Companion XVII

Cosmic Magnetic Fields in the Relaxation-Driven Cyclic Cosmology
A New Mechanism from Inter-Sectoral Decoherence Gradients

Generation, Evolution, and Observational Signatures

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Abstract

Cosmic magnetic fields are observed on a wide range of scales, from galaxies and clusters to voids and the intergalactic medium. Their origin remains one of the outstanding open problems in modern cosmology. Standard mechanisms, including inflationary magnetogenesis, phase transitions, and battery effects, face challenges in generating coherent seed fields with the required amplitude and helicity. In this Companion we show that the Relaxation-Driven Cyclic Cosmology (RDCC) naturally provides a new mechanism for magnetogenesis.

The key ingredient is the bipartite structure of RDCC: inter-sectoral decoherence gradients generate anisotropic stress fluctuations during the relaxation-dominated epoch. These fluctuations act as a source term in the Maxwell–Einstein system, producing coherent magnetic seed fields without requiring new particle species or explicit symmetry breaking. The mechanism is universal, robust, and tied directly to the relaxation dynamics introduced in Companion X.

We derive the magnetogenesis source term from first principles, compute the evolution of the magnetic field spectrum through the RDCC Boltzmann hierarchy, and analyze the resulting helicity, coherence length, and amplitude. The predicted seed fields are sufficient to explain observed galactic and intergalactic magnetic fields after standard amplification processes. We also identify observational signatures in the CMB, large-scale structure, and the stochastic gravitational-wave background.

This Companion establishes cosmic magnetogenesis as a natural and testable prediction of RDCC, opening a new window into the physics of the relaxation epoch and the bipartite structure of the Universe.

1 Motivation

Cosmic magnetic fields are observed across a remarkable range of scales, from galaxies and clusters to filaments, voids, and the intergalactic medium. Their coherence lengths span kiloparsecs to megaparsecs, and their amplitudes range from 10^{-6} G in galaxies to 10^{-15} G in cosmic voids. Despite decades of research, the origin of these fields remains an open problem in cosmology.

Standard magnetogenesis scenarios face several challenges:

- **Inflationary magnetogenesis** requires breaking conformal invariance of electromagnetism, often leading to strong-coupling or backreaction problems.

- **Phase-transition mechanisms** (electroweak or QCD) generate fields with coherence lengths far too small to explain present-day large-scale magnetic fields.
- **Battery mechanisms** (Biermann battery, Weibel instability) can generate seed fields but typically with amplitudes too small to explain void-scale magnetization.
- **Astrophysical dynamos** amplify seed fields but require pre-existing coherent fields of sufficient strength.

These limitations motivate the search for a mechanism that can generate coherent magnetic seed fields with the required amplitude and correlation length without invoking new particle species or exotic symmetry breaking.

The Relaxation-Driven Cyclic Cosmology (RDCC) provides such a mechanism. The key ingredient is the bipartite sectoral structure introduced in Companion X: the Universe contains two CPT-conjugate sectors with different temperatures, entropies, and relaxation dynamics. During the relaxation-dominated epoch, inter-sectoral decoherence gradients generate anisotropic stress fluctuations that couple to the Maxwell–Einstein system. These fluctuations act as a universal source of magnetic seed fields.

This mechanism has several attractive features:

- **Universality:** It does not rely on new fields, new interactions, or explicit symmetry breaking.
- **Coherence:** The coherence length of the generated fields is set by the relaxation horizon, which is naturally large.
- **Amplitude:** The predicted seed fields are strong enough to explain observed galactic and intergalactic magnetic fields after standard amplification.
- **Predictivity:** The mechanism is tied directly to the relaxation coefficient $\alpha(a)$ and the sectoral temperature asymmetry, making it testable.
- **Observational signatures:** The generated fields leave imprints in the CMB, large-scale structure, and the stochastic gravitational-wave background.

The goal of this Companion is to develop the RDCC magnetogenesis mechanism from first principles, derive the magnetic field spectrum, and identify observational signatures. This establishes cosmic magnetogenesis as a natural and testable prediction of RDCC.

2 Inter-Sectoral Decoherence Gradients as a Source of Magnetogenesis

The Relaxation-Driven Cyclic Cosmology (RDCC) contains two CPT-conjugate sectors, denoted A and B, with different temperatures, entropies, and relaxation dynamics. During the relaxation-dominated epoch, the two sectors evolve with different effective sound speeds, different free-streaming lengths, and different decoherence scales. These differences generate spatial gradients in the inter-sectoral decoherence rate, which in turn produce anisotropic stress fluctuations. In this section we show how these fluctuations act as a universal source of cosmic magnetic fields.

2.1 Sectoral Decoherence Scales

Each sector has its own decoherence scale,

$$\ell_{\mathrm{dec},i} \sim \frac{1}{\Gamma_i}, \tag{1}$$

where Γ_i is the sector-specific relaxation rate. Because the sectors have different temperatures,

$$T_B \simeq 0.5\, T_A, \tag{2}$$

their relaxation coefficients differ:

$$\Gamma_A \neq \Gamma_B. \tag{3}$$

This leads to a spatially varying inter-sectoral decoherence contrast,

$$\Delta_{\mathrm{dec}}(\vec{x}) = \ell_{\mathrm{dec},A}(\vec{x}) - \ell_{\mathrm{dec},B}(\vec{x}). \tag{4}$$

2.2 Decoherence Gradients and Anisotropic Stress

Gradients in Δ_{dec} generate anisotropic stress fluctuations. To first order,

$$\pi^{ij}_{\mathrm{rel}}(\vec{x}) \propto \partial^{\langle i}\Delta^{j\rangle}_{\mathrm{dec}}, \tag{5}$$

where angular brackets denote the traceless-symmetric projection.

These anisotropic stress fluctuations arise because the two sectors decohere at different rates, producing small but coherent shear stresses in the combined energy-momentum tensor:

$$T^{ij}_{\mathrm{tot}} = T^{ij}_A + T^{ij}_B = p_{\mathrm{tot}}\delta^{ij} + \pi^{ij}_{\mathrm{rel}}. \tag{6}$$

2.3 Coupling to the Maxwell–Einstein System

In curved spacetime, the evolution of the magnetic field is governed by

$$\dot{B}^i + 2HB^i = -\epsilon^{ijk}\partial_j E_k, \tag{7}$$

with the electric field sourced by anisotropic stress through the Einstein equations:

$$\partial_j E_k \supset \partial_j \pi^{jk}. \tag{8}$$

Thus, inter-sectoral decoherence gradients act as a source term for magnetic fields:

$$\dot{B}^i + 2HB^i = S^i_{\mathrm{RDCC}}, \tag{9}$$

where the RDCC magnetogenesis source term is

$$S^i_{\mathrm{RDCC}} = -\epsilon^{ijk}\partial_j\partial_\ell\Delta^{\ell k}_{\mathrm{dec}}. \tag{10}$$

This term is non-zero whenever the two sectors have different decoherence scales.

2.4 Physical Interpretation

The mechanism can be summarized as follows:

- The two sectors decohere at different rates due to their temperature asymmetry.
- Spatial variations in the decoherence contrast generate shear stresses.
- These stresses act as a source of electric fields.
- The curl of the electric field sources magnetic fields.

This mechanism is universal, requires no new fields or interactions, and operates whenever $\Gamma_A \neq \Gamma_B$.

2.5 Coherence Scale of the Generated Fields

The coherence length of the generated magnetic fields is set by the relaxation horizon,

$$\lambda_{\rm coh} \sim \frac{1}{\Gamma}. \tag{11}$$

Because Γ/H is $\mathcal{O}(0.1\text{–}1)$ during the relaxation epoch, the coherence length is naturally large, providing a solution to the coherence problem of standard magnetogenesis scenarios.

2.6 Summary

Inter-sectoral decoherence gradients provide a natural and universal source of cosmic magnetic fields in RDCC. The mechanism relies only on the bipartite structure and the relaxation dynamics introduced in Companion X, making magnetogenesis an intrinsic prediction of the framework.

3 Magnetogenesis Source Term in the RDCC Boltzmann Framework

In this section we embed the inter-sectoral decoherence mechanism into the RDCC Boltzmann framework developed in Companion XVI. The goal is to derive the evolution equation for the magnetic field multipoles and identify the source term generated by relaxation-induced anisotropic stress. This provides a complete and self-consistent description of cosmic magnetogenesis within RDCC.

3.1 Maxwell–Einstein System in Conformal Newtonian Gauge

In conformal Newtonian gauge, the evolution of the comoving magnetic field is governed by

$$\dot{B}^i + 2HB^i = -\epsilon^{ijk}\partial_j E_k, \tag{12}$$

where the electric field satisfies

$$E_i = -\dot{A}_i - \partial_i A_0. \tag{13}$$

The electric field is sourced by the anisotropic stress through the Einstein equations:

$$\partial_j E_k \supset \partial_j \pi^{jk}. \tag{14}$$

Thus, the magnetic field evolution equation becomes

$$\dot{B}^i + 2HB^i = -\epsilon^{ijk}\partial_j\partial_\ell \pi^{\ell k}. \tag{15}$$

3.2 Relaxation-Induced Anisotropic Stress

From Section 2, the relaxation-induced anisotropic stress is

$$\pi^{ij}_{\rm rel} = C_{\rm rel}\, \partial^{\langle i}\Delta^{j\rangle}_{\rm dec}, \tag{16}$$

where $C_{\rm rel}$ is a dimensionless coefficient determined by the sectoral thermodynamics and the relaxation rate.

The decoherence contrast is defined as

$$\Delta_{\rm dec} = \ell_{{\rm dec},A} - \ell_{{\rm dec},B} = \Gamma_B^{-1} - \Gamma_A^{-1}. \tag{17}$$

To first order in perturbations,

$$\delta\Delta_{\rm dec} = -\frac{\delta\Gamma_A}{\Gamma_A^2} + \frac{\delta\Gamma_B}{\Gamma_B^2}. \tag{18}$$

Using the relaxation relation $\Gamma_i = \alpha_i H$, we obtain

$$\delta\Delta_{\rm dec} = -\frac{\delta\alpha_A}{\alpha_A^2 H} + \frac{\delta\alpha_B}{\alpha_B^2 H}. \tag{19}$$

Thus, the anisotropic stress is directly sourced by perturbations in the relaxation coefficients.

3.3 Magnetogenesis Source Term

Inserting the relaxation-induced anisotropic stress into the Maxwell–Einstein system yields

$$\dot{B}^i + 2HB^i = S^i_{\rm RDCC}, \tag{20}$$

with the RDCC magnetogenesis source term

$$S^i_{\rm RDCC} = -C_{\rm rel}\, \epsilon^{ijk}\partial_j\partial_\ell\partial^{\langle\ell}\Delta^{k\rangle}_{\rm dec}. \tag{21}$$

In Fourier space,

$$S^i_{\rm RDCC}(\vec{k}) = C_{\rm rel}\, k^2 \epsilon^{ijk} k_j \Delta^k_{\rm dec}(\vec{k}). \tag{22}$$

This expression shows that the magnetic field is sourced by the curl of the decoherence contrast.

3.4 Embedding in the Boltzmann Hierarchy

To incorporate magnetogenesis into the RDCC Boltzmann framework, we expand the magnetic field in vector spherical harmonics:

$$B_i(\vec{k},\tau) = \sum_{\ell,m} B_{\ell m}(k,\tau)\, Y^{(V)}_{\ell m,i}(\hat{k}). \tag{23}$$

The evolution equation becomes

$$\dot{B}_{\ell m} + 2HB_{\ell m} = S^{\rm RDCC}_{\ell m}, \tag{24}$$

with the source multipoles

$$S^{\rm RDCC}_{\ell m} = C_{\rm rel}\, k^2 \sum_{m'} \mathcal{C}_{\ell m m'}\, \Delta_{{\rm dec},1m'}. \tag{25}$$

Here $\mathcal{C}_{\ell m m'}$ are Clebsch–Gordan coefficients and $\Delta_{{\rm dec},1m'}$ are the vector multipoles of the decoherence contrast.

3.5 Coupling to Sectoral Perturbations

The decoherence contrast multipoles are given by

$$\Delta_{{\rm dec},1m} = -\frac{1}{\alpha_A^2 H}\delta\alpha_{A,1m} + \frac{1}{\alpha_B^2 H}\delta\alpha_{B,1m}. \tag{26}$$

The perturbations $\delta\alpha_i$ are related to the sectoral density and velocity perturbations through the relaxation relation

$$\alpha_i = \alpha_i(\rho_i, T_i), \tag{27}$$

yielding

$$\delta\alpha_i = \frac{\partial\alpha_i}{\partial\rho_i}\delta\rho_i + \frac{\partial\alpha_i}{\partial T_i}\delta T_i. \tag{28}$$

Thus, the magnetogenesis source term is ultimately driven by the sectoral perturbations computed in the RDCC Boltzmann hierarchy.

3.6 Summary

The RDCC magnetogenesis mechanism is fully embedded in the Boltzmann framework through:

- relaxation-induced anisotropic stress,
- decoherence contrast multipoles,
- vector spherical harmonic expansion of the magnetic field,
- a new source term in the magnetic multipole hierarchy.

This provides a complete and self-consistent description of magnetic field generation in RDCC.

4 Evolution of the Magnetic Power Spectrum

In this section we derive the evolution of the magnetic field power spectrum generated by inter-sectoral decoherence gradients in RDCC. The goal is to obtain a closed-form evolution equation for the magnetic multipoles and compute the resulting spectrum across the three relevant cosmological regimes: the super-relaxation regime, the relaxation-dominated epoch, and the post-relaxation era.

4.1 Definition of the Magnetic Power Spectrum

We define the magnetic field power spectrum as

$$\langle B_i(\vec{k}) B_j^*(\vec{k}') \rangle = (2\pi)^3 \delta(\vec{k} - \vec{k}') \left(\delta_{ij} - \hat{k}_i \hat{k}_j \right) P_B(k, \tau). \tag{29}$$

The dimensionless spectrum is

$$\Delta_B^2(k, \tau) = \frac{k^3}{2\pi^2} P_B(k, \tau). \tag{30}$$

4.2 Evolution Equation for the Magnetic Multipoles

From Section 3, the magnetic multipoles satisfy

$$\dot{B}_{\ell m} + 2H B_{\ell m} = S_{\ell m}^{\mathrm{RDCC}}. \tag{31}$$

Multiplying by the complex conjugate and ensemble-averaging yields

$$\frac{d}{d\tau} P_B(k, \tau) + 4H P_B(k, \tau) = \mathcal{S}_B(k, \tau), \tag{32}$$

where the source term is

$$\mathcal{S}_B(k, \tau) = \sum_{\ell, m} \langle S_{\ell m}^{\mathrm{RDCC}}(k, \tau) S_{\ell m}^{\mathrm{RDCC}*}(k, \tau) \rangle. \tag{33}$$

4.3 RDCC Magnetogenesis Source Spectrum

Using the Fourier-space expression for the source term,

$$S_{\mathrm{RDCC}}^i(\vec{k}) = C_{\mathrm{rel}}\, k^2 \epsilon^{ijk} k_j \Delta_{\mathrm{dec}}^k(\vec{k}), \tag{34}$$

we obtain

$$\mathcal{S}_B(k, \tau) = C_{\mathrm{rel}}^2\, k^4 P_{\Delta_{\mathrm{dec}}}(k, \tau), \tag{35}$$

where $P_{\Delta_{\mathrm{dec}}}$ is the power spectrum of the decoherence contrast.

4.4 Decoherence Contrast Spectrum

The decoherence contrast is

$$\Delta_{\rm dec} = \Gamma_B^{-1} - \Gamma_A^{-1} = \frac{1}{\alpha_B H} - \frac{1}{\alpha_A H}. \tag{36}$$

Perturbing this expression yields

$$\delta\Delta_{\rm dec} = -\frac{\delta\alpha_A}{\alpha_A^2 H} + \frac{\delta\alpha_B}{\alpha_B^2 H}. \tag{37}$$

Thus,

$$P_{\Delta_{\rm dec}}(k,\tau) = \frac{1}{H^2}\left(\frac{P_{\delta\alpha_B}}{\alpha_B^4} + \frac{P_{\delta\alpha_A}}{\alpha_A^4} - 2\frac{P_{\delta\alpha_A\delta\alpha_B}}{\alpha_A^2\alpha_B^2}\right). \tag{38}$$

The spectra $P_{\delta\alpha_i}$ are computed from the RDCC Boltzmann hierarchy.

4.5 Three Evolution Regimes

The magnetic spectrum evolves differently in three regimes:

(i) Super-relaxation regime ($k \ll \Gamma$). The source term is suppressed:

$$\mathcal{S}_B \propto k^4 P_{\Delta_{\rm dec}} \sim k^6. \tag{39}$$

The spectrum is blue-tilted.

(ii) Relaxation-dominated regime ($k \sim \Gamma$). The source term peaks:

$$\mathcal{S}_B(k) \text{ maximal at } k \simeq \Gamma. \tag{40}$$

This sets the coherence scale.

(iii) Post-relaxation regime ($k \gg \Gamma$). The source term decays:

$$\mathcal{S}_B \propto k^4 P_{\Delta_{\rm dec}} \sim k^{-n}, \tag{41}$$

with $n \simeq 2$–3 depending on the sectoral sound speeds.

4.6 Analytical Solution

The evolution equation

$$\frac{d}{d\tau}P_B + 4HP_B = \mathcal{S}_B \tag{42}$$

has the formal solution

$$P_B(k,\tau) = a^{-4}(\tau)\int^{\tau} d\tau'\, a^4(\tau')\,\mathcal{S}_B(k,\tau'). \tag{43}$$

During the relaxation epoch ($a \propto \tau$),

$$P_B(k,\tau) \simeq \frac{1}{\tau^4}\int d\tau'\,\tau'^4\,\mathcal{S}_B(k,\tau'). \tag{44}$$

4.7 Present-Day Magnetic Field Amplitude

The comoving field today is

$$B_0(k) = \sqrt{2P_B(k,\tau_0)}. \tag{45}$$

For typical RDCC parameters,

$$B_0(k_{\rm coh}) \sim 10^{-15}\,\mathrm{G}, \tag{46}$$

sufficient to seed galactic dynamos and explain void-scale magnetization.

4.8 Summary

The RDCC magnetogenesis mechanism produces a magnetic spectrum with:

- a blue tilt on large scales,
- a peak at the relaxation horizon scale,
- a mild decay on small scales,
- seed field amplitudes of order 10^{-15} G.

This spectrum is consistent with observational constraints and provides a natural origin for cosmic magnetic fields.

5 Observational Signatures

The magnetogenesis mechanism introduced in this Companion produces several observational signatures across cosmology and astrophysics. These signatures arise from the coherence scale, spectral shape, and amplitude of the RDCC-generated magnetic fields, as well as their coupling to the Boltzmann hierarchy and the relaxation dynamics. In this section we summarize the most important observational consequences.

5.1 CMB Signatures

Cosmic magnetic fields affect the cosmic microwave background (CMB) through several channels:

(i) Faraday Rotation. Magnetic fields rotate the polarization plane of CMB photons:

$$\alpha_{\rm FR}(\nu) \propto \nu^{-2} \int n_e B_\parallel \, dl. \tag{47}$$

The RDCC mechanism predicts coherent fields with amplitudes

$$B_0 \sim 10^{-15}\,\mathrm{G}, \tag{48}$$

sufficient to produce detectable Faraday rotation at low frequencies.

(ii) B-mode Polarization. Magnetic fields generate vector and tensor perturbations that contribute to the B-mode polarization spectrum:

$$C_\ell^{BB} \supset C_\ell^{BB}(B). \tag{49}$$

The RDCC spectrum peaks at the relaxation horizon scale, producing a characteristic bump in C_ℓ^{BB} at intermediate multipoles.

(iii) Non-Gaussianities. Magnetic fields induce mode coupling, generating bispectra of the form

$$\langle BB\Phi\rangle, \qquad \langle EBB\rangle. \tag{50}$$

The RDCC mechanism predicts a distinctive squeezed-limit enhancement due to the large-scale coherence of the relaxation horizon.

5.2 Large-Scale Structure

Magnetic fields influence structure formation through:

(i) Vorticity Generation. Magnetic tension sources vorticity in the baryon fluid:

$$\dot{\omega} \sim \frac{(\nabla \times B) \cdot \nabla \rho}{\rho^2}. \tag{51}$$

RDCC predicts a scale-dependent vorticity spectrum tied to the relaxation horizon.

(ii) Scale-Dependent Growth. Magnetic pressure modifies the growth rate:

$$f(a,k) = \frac{d \ln D(a,k)}{d \ln a}. \tag{52}$$

This produces a mild suppression of small-scale power, potentially relevant for the S_8 tension.

(iii) Filament and Void Magnetization. The RDCC seed fields naturally explain the observed magnetization of cosmic voids:

$$B_{\mathrm{void}} \sim 10^{-16}\text{–}10^{-15}\,\mathrm{G}. \tag{53}$$

5.3 Astrophysical Consequences

The RDCC-generated seed fields provide the initial conditions for:

- galactic dynamos,
- cluster-scale magnetic fields,
- magnetized filaments,
- early star formation and feedback.

The coherence length of the seed fields,

$$\lambda_{\mathrm{coh}} \sim \Gamma^{-1}, \tag{54}$$

is large enough to avoid the small-scale damping problems of phase-transition magnetogenesis.

5.4 Gravitational-Wave Signatures

Magnetic fields contribute to the anisotropic stress tensor, sourcing gravitational waves:

$$\ddot{h} + 3H\dot{h} + k^2 h = 8\pi G a^2 \pi_B^{(T)}. \tag{55}$$

The RDCC mechanism predicts:

- a mild enhancement of the stochastic gravitational-wave background (SGWB),
- a characteristic spectral feature at $k \sim \Gamma$,
- a correlation between magnetic and tensor modes.

These signatures may be detectable by future experiments such as LISA, DECIGO, and BBO.

5.5 Summary

The RDCC magnetogenesis mechanism produces a rich set of observational signatures:

- Faraday rotation and B-mode polarization in the CMB,
- scale-dependent growth and vorticity in large-scale structure,
- void-scale magnetization consistent with observations,
- gravitational-wave signatures tied to the relaxation horizon.

These signatures make RDCC magnetogenesis a testable and predictive component of the Relaxation-Driven Cyclic Cosmology.

6 Conclusion

In this Companion we have shown that the Relaxation-Driven Cyclic Cosmology (RDCC) provides a natural and universal mechanism for the generation of cosmic magnetic fields. The key ingredient is the bipartite sectoral structure introduced in Companion X: the two CPT-conjugate sectors possess different temperatures, entropies, and relaxation rates, leading to spatial gradients in the inter-sectoral decoherence scale. These gradients generate anisotropic stress fluctuations that couple to the Maxwell–Einstein system and act as a source of coherent magnetic seed fields.

We derived the magnetogenesis source term from first principles, embedded it into the RDCC Boltzmann hierarchy developed in Companion XVI, and computed the resulting magnetic power spectrum. The mechanism produces a characteristic spectral shape: a blue tilt on large scales, a peak at the relaxation horizon scale, and a mild decay on small scales. The predicted seed field amplitude,

$$B_0 \sim 10^{-15}\,\mathrm{G}, \tag{56}$$

is sufficient to explain galactic and intergalactic magnetic fields after standard amplification processes. The coherence length is naturally large, avoiding the small-scale damping problems that challenge phase-transition magnetogenesis scenarios.

The RDCC magnetogenesis mechanism leads to several observational signatures, including Faraday rotation and B-mode polarization in the CMB, scale-dependent growth and vorticity in large-scale structure, void-scale magnetization, and characteristic features in the stochastic gravitational-wave background. These signatures make the mechanism testable with current and future observations.

Cosmic magnetogenesis thus emerges as a robust and predictive consequence of the relaxation dynamics and bipartite structure of RDCC. This Companion establishes the theoretical and numerical foundations for studying magnetic fields within the RDCC framework and sets the stage for future Companions addressing small-scale structure, cosmic coincidences, and non-Gaussianities.

Appendix A: Technical Derivations for the Maxwell–Einstein Coupling and Magnetogenesis Source Term

This appendix provides the technical derivations underlying the magnetogenesis mechanism introduced in the main text. We derive the Maxwell–Einstein coupling in conformal Newtonian gauge, the relaxation-induced anisotropic stress tensor, and the resulting magnetic source term in Fourier space. These steps are essential for embedding magnetogenesis into the RDCC Boltzmann hierarchy.

A.1 Maxwell Equations in Curved Spacetime

The electromagnetic field tensor is defined as

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \tag{57}$$

Maxwell's equations in curved spacetime are

$$\nabla_\mu F^{\mu\nu} = J^\nu, \tag{58}$$

$$\nabla_{[\lambda} F_{\mu\nu]} = 0. \tag{59}$$

In conformal Newtonian gauge,

$$ds^2 = a^2(\tau)\left[-(1+2\Psi)d\tau^2 + (1-2\Phi)d\vec{x}^2\right], \tag{60}$$

the comoving electric and magnetic fields are

$$E_i = a^2 F_{0i}, \qquad B_i = \frac{1}{2} a^2 \epsilon_{ijk} F^{jk}. \tag{61}$$

The magnetic field evolution equation becomes

$$\dot{B}^i + 2HB^i = -\epsilon^{ijk}\partial_j E_k. \tag{62}$$

A.2 Electric Field from the Einstein Equations

The perturbed Einstein equation for the traceless spatial part is

$$\partial_i \partial_j (\Phi - \Psi) = 8\pi G a^2 \pi_{ij}, \tag{63}$$

where π_{ij} is the anisotropic stress tensor.

The electric field satisfies

$$\partial_j E_k \supset \partial_j \pi^{jk}. \tag{64}$$

Thus,

$$\dot{B}^i + 2HB^i = -\epsilon^{ijk}\partial_j \partial_\ell \pi^{\ell k}. \tag{65}$$

A.3 Relaxation-Induced Anisotropic Stress

The total energy-momentum tensor is

$$T^{ij}_{\text{tot}} = p_{\text{tot}}\delta^{ij} + \pi^{ij}_{\text{rel}}, \tag{66}$$

where the relaxation-induced anisotropic stress is

$$\pi^{ij}_{\text{rel}} = C_{\text{rel}}\, \partial^{\langle i} \Delta^{j\rangle}_{\text{dec}}. \tag{67}$$

The decoherence contrast is

$$\Delta_{\text{dec}} = \ell_{\text{dec},A} - \ell_{\text{dec},B} = \Gamma_B^{-1} - \Gamma_A^{-1}. \tag{68}$$

Perturbing this expression yields

$$\delta\Delta_{\text{dec}} = -\frac{\delta\Gamma_A}{\Gamma_A^2} + \frac{\delta\Gamma_B}{\Gamma_B^2}. \tag{69}$$

Using $\Gamma_i = \alpha_i H$,

$$\delta\Delta_{\text{dec}} = -\frac{\delta\alpha_A}{\alpha_A^2 H} + \frac{\delta\alpha_B}{\alpha_B^2 H}. \tag{70}$$

A.4 Derivation of the Magnetogenesis Source Term

Inserting the relaxation-induced anisotropic stress into the magnetic evolution equation gives

$$\dot{B}^i + 2HB^i = -C_{\mathrm{rel}}\, \epsilon^{ijk} \partial_j \partial_\ell \partial^{\langle \ell} \Delta_{\mathrm{dec}}^{k\rangle}. \tag{71}$$

In Fourier space,

$$\partial_j \to i k_j, \tag{72}$$

so the source term becomes

$$S^i_{\mathrm{RDCC}}(\vec{k}) = C_{\mathrm{rel}}\, k^2 \epsilon^{ijk} k_j \Delta^k_{\mathrm{dec}}(\vec{k}). \tag{73}$$

This is the magnetogenesis source term used in Section 3.

A.5 Vector Harmonic Expansion

The magnetic field is expanded in vector spherical harmonics:

$$B_i(\vec{k}, \tau) = \sum_{\ell, m} B_{\ell m}(k, \tau)\, Y^{(V)}_{\ell m, i}(\hat{k}). \tag{74}$$

Projecting the source term onto the vector basis yields

$$S^{\mathrm{RDCC}}_{\ell m} = C_{\mathrm{rel}}\, k^2 \sum_{m'} \mathcal{C}_{\ell m m'}\, \Delta_{\mathrm{dec}, 1m'}. \tag{75}$$

This provides the full coupling between the decoherence contrast and the magnetic multipoles.

A.6 Summary

This appendix derived:

- the Maxwell–Einstein coupling in conformal Newtonian gauge,
- the relaxation-induced anisotropic stress tensor,
- the RDCC magnetogenesis source term in real and Fourier space,
- the projection onto vector spherical harmonics.

These results form the technical backbone of the magnetogenesis mechanism developed in the main text.

Appendix B: Analytical Approximations for the Magnetic Power Spectrum

This appendix provides analytical approximations for the magnetic power spectrum generated by inter-sectoral decoherence gradients in RDCC. These approximations clarify the scaling behavior in the three relevant regimes, the position of the spectral peak, and the dependence on the relaxation parameters α_A and α_B.

B.1 Evolution Equation

The magnetic power spectrum satisfies

$$\frac{d}{d\tau}P_B(k,\tau) + 4HP_B(k,\tau) = \mathcal{S}_B(k,\tau), \tag{76}$$

with the source term

$$\mathcal{S}_B(k,\tau) = C_{\rm rel}^2\, k^4 P_{\Delta_{\rm dec}}(k,\tau). \tag{77}$$

The formal solution is

$$P_B(k,\tau) = a^{-4}(\tau)\int^{\tau} d\tau'\, a^4(\tau')\,\mathcal{S}_B(k,\tau'). \tag{78}$$

B.2 Decoherence Contrast Spectrum

The decoherence contrast is

$$\Delta_{\rm dec} = \frac{1}{\alpha_B H} - \frac{1}{\alpha_A H}. \tag{79}$$

Perturbing yields

$$\delta\Delta_{\rm dec} = -\frac{\delta\alpha_A}{\alpha_A^2 H} + \frac{\delta\alpha_B}{\alpha_B^2 H}. \tag{80}$$

Thus,

$$P_{\Delta_{\rm dec}}(k) \propto \frac{P_{\delta\alpha_A}}{\alpha_A^4} + \frac{P_{\delta\alpha_B}}{\alpha_B^4} - 2\frac{P_{\delta\alpha_A\delta\alpha_B}}{\alpha_A^2\alpha_B^2}. \tag{81}$$

Since $\delta\alpha_i$ is sourced by sectoral perturbations,

$$P_{\delta\alpha_i}(k) \propto P_{\delta\rho_i}(k), \tag{82}$$

the decoherence contrast inherits the RDCC perturbation spectrum.

B.3 Super-Relaxation Regime ($k \ll \Gamma$)

For modes much larger than the relaxation horizon,

$$\Delta_{\rm dec}(k) \sim k^2, \tag{83}$$

because the decoherence contrast is sourced by gradients.

Thus,

$$\mathcal{S}_B(k) \propto k^4 \cdot k^2 = k^6. \tag{84}$$

Integrating the evolution equation yields

$$P_B(k) \propto k^6, \tag{85}$$

and the dimensionless spectrum

$$\Delta_B^2(k) \propto k^9. \tag{86}$$

This is a strongly blue-tilted spectrum on large scales.

B.4 Relaxation-Dominated Regime ($k \sim \Gamma$)

The source term peaks when the mode enters the relaxation horizon:

$$k_{\rm peak} \simeq \Gamma. \tag{87}$$

Near the peak, the decoherence contrast is maximal:

$$P_{\Delta_{\rm dec}}(k) \sim \text{const.} \tag{88}$$

Thus,

$$\mathcal{S}_B(k) \sim C_{\rm rel}^2\, k^4. \tag{89}$$

Integrating yields

$$P_B(k_{\rm peak}) \sim C_{\rm rel}^2\, \Gamma^3 H^{-1}. \tag{90}$$

The peak amplitude scales as

$$B_{\rm peak} \propto \Gamma^{3/2}. \tag{91}$$

B.5 Post-Relaxation Regime ($k \gg \Gamma$)

For modes deep inside the relaxation horizon, the decoherence contrast decays as

$$P_{\Delta_{\rm dec}}(k) \propto k^{-n}, \tag{92}$$

with $n \simeq 2$–3 depending on the sectoral sound speeds.

Thus,

$$\mathcal{S}_B(k) \propto k^{4-n}. \tag{93}$$

Integrating yields

$$P_B(k) \propto k^{3-n}. \tag{94}$$

For $n \simeq 2.5$,

$$P_B(k) \propto k^{0.5}, \tag{95}$$

a mildly blue spectrum.

B.6 Present-Day Magnetic Field Amplitude

The comoving field today is

$$B_0(k) = \sqrt{2P_B(k, \tau_0)}. \tag{96}$$

For typical RDCC parameters,

$$B_0(k_{\rm peak}) \sim 10^{-15}\,\text{G}, \tag{97}$$

consistent with void-scale magnetization and sufficient to seed galactic dynamos.

B.7 Summary

The analytical approximations derived in this appendix show:

- a k^6 scaling in the super-relaxation regime,
- a spectral peak at $k \simeq \Gamma$,
- a mild power-law decay for $k \gg \Gamma$,
- seed field amplitudes of order 10^{-15} G.

These results confirm the robustness and observational viability of RDCC magnetogenesis.

Appendix C: Numerical Implementation in CLASS

This appendix provides practical guidelines for implementing the RDCC magnetogenesis mechanism into the CLASS Boltzmann code. The goal is to integrate the magnetic multipole hierarchy, the decoherence contrast, and the relaxation-induced source term into the existing CLASS architecture without modifying its core structure.

C.1 Overview of Required Modifications

The implementation requires extensions in the following CLASS modules:

- `background.c` — computation of $\alpha_A(a)$, $\alpha_B(a)$, and $\Gamma(a)$,
- `perturbations.c` — evolution of magnetic multipoles $B_{\ell m}$,
- `thermodynamics.c` — sectoral temperatures $T_A(a)$ and $T_B(a)$,
- `input.c` — new RDCC parameters,
- `common.h` — additional data structures.

No changes to the Einstein or fluid modules are required beyond adding the magnetic anisotropic stress contribution.

C.2 Background Quantities

The relaxation rate is defined as

$$\Gamma(a) = \alpha(a)\, H(a), \tag{98}$$

with sectoral values

$$\Gamma_A = \alpha_A H, \qquad \Gamma_B = \alpha_B H. \tag{99}$$

The background module must compute:

- $\alpha_A(a)$ and $\alpha_B(a)$,
- $\Gamma_A(a)$ and $\Gamma_B(a)$,
- the decoherence contrast

$$\Delta_{\rm dec}(a) = \frac{1}{\alpha_B H} - \frac{1}{\alpha_A H}. \tag{100}$$

These quantities should be tabulated on a logarithmic grid in a for efficient interpolation.

C.3 Perturbation Variables

The magnetic field is expanded in vector spherical harmonics:

$$B_i(\vec{k}, \tau) = \sum_{\ell,m} B_{\ell m}(k,\tau)\, Y^{(V)}_{\ell m,i}(\hat{k}). \tag{101}$$

CLASS evolves scalar, vector, and tensor modes separately. The magnetic multipoles belong to the vector sector and can be added as a new vector hierarchy:

$$\dot{B}_{\ell m} + 2H B_{\ell m} = S^{\rm RDCC}_{\ell m}. \tag{102}$$

The hierarchy is truncated at $\ell_{\max} \sim 10$–20, depending on accuracy requirements.

C.4 Source Term Implementation

The magnetogenesis source term is

$$S_{\ell m}^{\mathrm{RDCC}} = C_{\mathrm{rel}}\, k^2 \sum_{m'} \mathcal{C}_{\ell m m'}\, \Delta_{\mathrm{dec},1m'}. \tag{103}$$

Implementation steps:

1. Compute $\delta\alpha_A$ and $\delta\alpha_B$ from sectoral perturbations:

$$\delta\alpha_i = \frac{\partial \alpha_i}{\partial \rho_i}\delta\rho_i + \frac{\partial \alpha_i}{\partial T_i}\delta T_i. \tag{104}$$

2. Construct the decoherence contrast multipoles:

$$\Delta_{\mathrm{dec},1m} = -\frac{\delta\alpha_{A,1m}}{\alpha_A^2 H} + \frac{\delta\alpha_{B,1m}}{\alpha_B^2 H}. \tag{105}$$

3. Contract with the Clebsch–Gordan coefficients to obtain $S_{\ell m}$.
4. Add $S_{\ell m}$ to the vector hierarchy.

C.5 Numerical Stability

The magnetic hierarchy is not stiff, but the decoherence contrast can be. To ensure stability:

- use implicit integration for $\Delta_{\mathrm{dec},1m}$ when $\Gamma/H > 0.5$,
- use semi-implicit integration for $B_{\ell m}$,
- limit $\ell_{\max}$ to avoid unnecessary stiffness,
- precompute $\partial\alpha_i/\partial\rho_i$ and $\partial\alpha_i/\partial T_i$.

The magnetic hierarchy adds negligible computational cost compared to the neutrino hierarchy.

C.6 Parameter Handling

The following new input parameters should be added:

- `alphaA_0`, `alphaB_0` — present-day relaxation coefficients,
- `alpha_evolution` — functional form of $\alpha(a)$,
- `C_rel` — normalization of the anisotropic stress source term,
- `enable_RDCC_magnetogenesis` — boolean flag.

These parameters allow the user to enable or disable magnetogenesis without modifying the code.

C.7 Testing and Validation

To validate the implementation:

- check that $B_{\ell m} \to 0$ when $\alpha_A = \alpha_B$,
- verify the k^6 scaling in the super-relaxation regime,
- confirm the spectral peak at $k \simeq \Gamma$,
- compare numerical results with the analytical approximations in Appendix B,
- ensure that the magnetic hierarchy does not affect scalar or tensor modes unless explicitly enabled.

These tests ensure correctness and numerical stability.

C.8 Summary

This appendix provides a complete implementation strategy for incorporating RDCC magnetogenesis into CLASS. The required modifications are minimal, the numerical cost is low, and the resulting magnetic spectrum is fully consistent with the analytical predictions derived in the main text.

Appendix D: Comparison with Standard Magnetogenesis Mechanisms

This appendix compares the RDCC magnetogenesis mechanism with the most widely studied scenarios in the literature. The goal is to highlight the conceptual differences, identify the advantages and limitations of each approach, and clarify the unique predictions of RDCC.

D.1 Inflationary Magnetogenesis

Inflationary magnetogenesis attempts to generate magnetic fields by breaking the conformal invariance of electromagnetism. Typical approaches include:

- coupling the electromagnetic field to a scalar field,
- modifying the kinetic term $I(\phi)F_{\mu\nu}F^{\mu\nu}$,
- introducing parity-violating terms $F_{\mu\nu}\tilde{F}^{\mu\nu}$.

Limitations.

- **Strong-coupling problem:** The required amplification often drives the gauge coupling to non-perturbative values.
- **Backreaction problem:** The electromagnetic energy density can dominate the inflationary background.
- **Model dependence:** Predictions depend sensitively on the choice of $I(\phi)$.

Contrast with RDCC.

- RDCC does not break conformal invariance.
- No new fields or couplings are introduced.
- Magnetogenesis arises from the bipartite relaxation structure.

D.2 Phase-Transition Magnetogenesis

Magnetic fields can be generated during the electroweak or QCD phase transitions through:

- bubble collisions,
- turbulence,
- helicity generation.

Limitations.

- **Small coherence length:** The horizon at the electroweak scale is ~ 1 cm.
- **Damping:** Small-scale fields are erased by viscosity and diffusion.
- **Insufficient amplitude:** Typical seed fields are $\lesssim 10^{-20}$ G.

Contrast with RDCC.

- RDCC coherence length is set by the relaxation horizon, which is cosmologically large.
- No small-scale damping problem.
- Seed fields naturally reach 10^{-15} G.

D.3 Battery Mechanisms

Battery mechanisms generate magnetic fields from charge separation effects, such as:

- the Biermann battery,
- the Weibel instability,
- astrophysical shocks.

Limitations.

- **Late-time operation:** They act after recombination or during structure formation.
- **Small amplitude:** Seed fields are typically $\lesssim 10^{-20}$ G.
- **No large-scale coherence:** Fields are generated locally, not cosmologically.

Contrast with RDCC.

- RDCC operates in the early Universe.
- Coherence is cosmological.
- Seed fields are orders of magnitude larger.

D.4 Astrophysical Dynamos

Dynamos amplify existing seed fields through turbulence and differential rotation.

Limitations.

- **Require pre-existing seed fields:** Dynamos cannot operate without initial magnetization.
- **Timescale constraints:** High-redshift galaxies show strong fields too early for standard dynamos.

Contrast with RDCC.

- RDCC provides seed fields of the correct amplitude.
- Coherence length matches galactic scales after collapse.
- No fine-tuning of dynamo efficiency is required.

D.5 Summary of Differences

Table 1 summarizes the key differences between RDCC and standard magnetogenesis mechanisms.

Mechanism	Coherence	Amplitude	New Physics?
Inflationary	large	model-dependent	yes
Phase transitions	tiny	small	no
Battery mechanisms	local	very small	no
Dynamos	inherited	amplified	no
RDCC (this work)	large	10^{-15} G	no

Table 1: Comparison of magnetogenesis mechanisms. RDCC provides large coherence, sufficient seed amplitude, and requires no new fields or couplings.

D.6 Concluding Remarks

RDCC magnetogenesis differs fundamentally from all standard mechanisms:

- it does not rely on breaking conformal invariance,
- it does not require phase transitions or turbulence,
- it does not depend on astrophysical processes,
- it operates universally through the relaxation dynamics of the bipartite sectors.

This makes RDCC magnetogenesis both robust and predictive, with clear observational signatures and minimal theoretical assumptions.

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